

Case study: Internet Archive — public infrastructure footprint + recent-incident retrospective

CASE-2026-002 - status OPEN - priority HIGH

CASE NUMBER

CASE-2026-002

CUSTODIAN

Admin

TEAM

-

CASE OPENED

2026-05-20 20:24:48 UTC

REPORT GENERATED

2026-05-21 11:58:27 UTC

GENERATED BY

Admin

Forensic case opened from investigation Investigation: archive.org. Target: archive.org

CHAIN-OF-CUSTODY ATTESTATION

VERIFIED

CHAIN LENGTH
6 entries

HEAD HASH (SHA-256)
d410ef2a1f4c003fbfc2be02f7b4c1c49536d4a51df98317fb0a521b454fc4c0

GENESIS HASH
00

REPORT DIGEST (SHA-256)
94740589dae4beef18a26833e903cdad3d43ae466c1acde269674ad4679f70de

Evidence inventory

11 items - 0 verified or sealed - 0 backed by real file bytes on disk.

NAME	TYPE	STATUS	SIZE	SHA-256	OBJECT KEY	ADDED
Finding cmpehyhw - Audio subdomains are load-balanced endpoints for the Wayback Machine	document application/x-osint-finding	COLLECTED	-	1e60bc2a70c5... SHA-256	(metadata-only)	2026-05-20 20:24:48 UTC
Finding cmpehyhw - Analytics subdomain is self-hosted on the same stack	document application/x-osint-finding	COLLECTED	-	11b3c2e66c35... SHA-256	(metadata-only)	2026-05-20 20:24:48 UTC
Finding cmpehyhw - Archive.org operates a large, centrally-managed subdomain ecosystem	document application/x-osint-finding	COLLECTED	-	7f3217593831... SHA-256	(metadata-only)	2026-05-20 20:24:48 UTC
Finding cmpehw3f - Archive.org's public web presence is served via a modern React front-end	document application/x-osint-finding	COLLECTED	-	6ded214b9da9... SHA-256	(metadata-only)	2026-05-20 20:24:48 UTC
Finding cmpehw3f - Archive.org runs internal development and QA environments under subdomains	document application/x-osint-finding	COLLECTED	-	faa6f2295f2c... SHA-256	(metadata-only)	2026-05-20 20:24:48 UTC
Finding cmpehyhv - identity pipeline returned unstructured output	document application/x-osint-finding	COLLECTED	-	eaf8b5715dde... SHA-256	(metadata-only)	2026-05-20 20:24:48 UTC
Finding cmpehw3e - Archive.org hosts audio streaming subdomains	document application/x-osint-finding	COLLECTED	-	34c48f5b4939... SHA-256	(metadata-only)	2026-05-20 20:24:48 UTC
Finding cmpehw3e - Archive.org operates a large petabox CDN infrastructure	document application/x-osint-finding	COLLECTED	-	adc35531c032... SHA-256	(metadata-only)	2026-05-20 20:24:48 UTC
Finding cmpehw3e - Key Personnel Email Pattern	document application/x-osint-finding	COLLECTED	-	9d93bfa4e2b3... SHA-256	(metadata-only)	2026-05-20 20:24:48 UTC
Finding cmpehw3d - Domain Ownership	document application/x-osint-finding	COLLECTED	-	ce12a5acd7e8... SHA-256	(metadata-only)	2026-05-20 20:24:48 UTC
Finding cmpehw3d - Internet Archive Founder	document application/x-osint-finding	COLLECTED	-	c03d65846a18... SHA-256	(metadata-only)	2026-05-20 20:24:48 UTC

Findings

11 findings across 1 investigations.

Audio subdomains are load-balanced endpoints for the Wayback Machine

INFRASTRUCTURE PROBABLE PRIORITY: LOW - Investigation: archive.org

Multiple audioXX.archive.org subdomains resolve to Nginx 1.31.0 and return HTTP 302 redirects, typical of load-balanced content delivery nodes for media assets used by the Wayback Machine.

```
{
  "technique": "KAC",
  "inputs": [
    {
      "sourceId": "httpx_probe:audio22.archive.org",
      "summary": "HTTPS 302 redirect, server nginx/1.31.0.",
      "credibility": 5,
      "recencyDays": 0
    },
    {
      "sourceId": "httpx_probe:all.archive.org",
      "summary": "HTTPS 302 redirect, server nginx/1.31.0.",
      "credibility": 5,
      "recencyDays": 0
    }
  ],
  "reasoning": "Consistent redirect behavior across many audioXX subdomains indicates they are not distinct services but part of a CDN-style distribution for archived audio content.",
  "outputCandidates": [
    {
      "label": "CDN/media distribution nodes",
      "weight": 0.15,
      "disconfirmingEvidence": ["Unique content served per subdomain without redirects"]
    },
    {
      "label": "Separate services",
      "weight": 0.85,
      "disconfirmingEvidence": ["Distinct application signatures or ports per subdomain"]
    }
  ],
  "selected": 0
}
```

Analytics subdomain is self-hosted on the same stack

INFRASTRUCTURE PROBABLE PRIORITY: MEDIUM - Investigation: archive.org

The subdomain analytics.archive.org resolves to a live HTTPS service returning a 200 status with Nginx 1.31.0 and HSTS. No third-party analytics domain (e.g., google-analytics.com) is observed, implying the Internet Archive runs its own analytics platform.

```
{
  "technique": "Indicators",
  "inputs": [
    {
      "sourceId": "httpx_probe:analytics.archive.org",
      "summary": "HTTPS 200, server nginx/1.31.0, HSTS enabled.",
      "credibility": 5,
      "recencyDays": 0
    }
  ],
  "reasoning": "Self-hosted analytics would appear as a subdomain under the primary domain and share the same server technology. Absence of external analytics domains in page source (not fetched here) would further support this.",
  "outputCandidates": [
    {
      "label": "Self-hosted analytics",
      "weight": 0.2,
      "disconfirmingEvidence": ["Presence of external analytics script URLs in the page source"]
    },
    {
      "label": "Redirect to third-party service",
      "weight": 0.8,
      "disconfirmingEvidence": ["HTTP 302 to a non-archive.org analytics provider"]
    }
  ],
  "selected": 0
}
```

Archive.org operates a large, centrally-managed subdomain ecosystem

INFRASTRUCTURE CONFIRMED PRIORITY: HIGH - Investigation: archive.org

Passive subdomain enumeration returned 50 distinct subdomains, many of which resolve to live HTTPS services. HTTP probing shows most respond with Nginx server headers and similar TLS configurations, indicating they are served from a common infrastructure under the control of the Internet Archive.

```
{
  "technique": "QoIC",
  "inputs": [
    {
      "sourceId": "subfinder_domain:archive.org",
      "summary": "Returned 50 subdomains including all.archive.org, analytics.archive.org, ark.archive.org, etc.",
      "credibility": 5,
      "recencyDays": 0
    },
    {
      "sourceId": "httpx_probe:selected",
      "summary": "Probed 7 subdomains; 6 were alive, all returned Nginx server headers and similar TLS stacks.",
      "credibility": 5,
      "recencyDays": 0
    }
  ],
  "reasoning": "A high count of subdomains combined with uniform server technology strongly suggests a single administrative domain managing the infrastructure.",
  "outputCandidates": [
    {
      "label": "Centralized control",
      "weight": 0.1,
      "disconfirmingEvidence": ["Discovery of subdomains hosted on distinct cloud providers or with divergent server headers"]
    },
    {
      "label": "Fragmented control",
      "weight": 0.9,
      "disconfirmingEvidence": ["Evidence of separate ASNs or IP ranges for different subdomains"]
    }
  ],
  "selected": 0
}
```

identity pipeline returned unstructured output

IDENTITY UNVERIFIED PRIORITY: LOW - Investigation: archive.org

(no model output - adapter "openrouter" model "openai/gpt-oss-120b:free" produced no content)

```
{
  "error": "extractJson: no JSON opener in ",
  "raw_excerpt": ""
}
```

Archive.org's public web presence is served via a modern React front-end

INFRASTRUCTURE CONFIRMED PRIORITY: LOW - Investigation: archive.org

Fetching https://archive.org returned HTML that loads JavaScript modules (e.g., /offshoot_assets/index-DQEDXcHO.js) and includes a <app-root> custom element, indicating a single-page application architecture.

```
{
  "technique": "QoIC",
  "inputs": [
    {
      "sourceId": "http_fetch:https://archive.org",
      "summary": "HTML contains <app-root> and loads module scripts from /offshoot_assets/.",
      "credibility": 5,
      "recencyDays": 0
    }
  ],
  "reasoning": "The page structure directly shows the"
}
```

```
technology stack, confirming the use of a modern SPA framework for the public site.", "outputCandidates": [{"label": "SPA front-end confirmed", "weight": 0.05, "disconfirmingEvidence": ["A later fetch returns a static HTML page without JavaScript assets."]}], "selected": 0}
```

Archive.org runs internal development and QA environments under subdomains

INFRASTRUCTURE **PROBABLE** **PRIORITY: MEDIUM** - Investigation: archive.org

Subdomains such as dev.archive.org, qa-webservices.archive.org, qa-loc.archive.org, and ia-devo.dev.archive.org were found, revealing separate environments for development, testing, and quality assurance.

```
{"technique": "KAC", "inputs": [{"sourceId": "subfinder_domain:archive.org", "summary": "Presence of 'dev', 'qa', and 'devo' prefixes in subdomains.", "credibility": 3, "recencyDays": 0}], "reasoning": "These prefixes are commonly used for internal staging environments. Their existence suggests a structured development pipeline, which may expose misconfigurations if publicly reachable.", "outputCandidates": [{"label": "Staging environments exposed", "weight": 0.4, "disconfirmingEvidence": ["All staging subdomains resolve to private IP ranges or return access-denied responses."]}], "selected": 0}
```

Archive.org hosts audio streaming subdomains

INFRASTRUCTURE **PROBABLE** **PRIORITY: MEDIUM** - Investigation: archive.org

Subdomains such as audio09.archive.org, audio11-bu.archive.org, audio27.archive.org, and audio18.archive.org were enumerated, indicating dedicated audio streaming endpoints within the IA infrastructure.

```
{"technique": "Indicators", "inputs": [{"sourceId": "subfinder_domain:archive.org", "summary": "Multiple 'audioXX.archive.org' hostnames discovered.", "credibility": 3, "recencyDays": 0}], "reasoning": "The naming convention suggests purpose-built audio delivery services. Confirmation would come from HTTP probing showing audio MIME types or streaming headers.", "outputCandidates": [{"label": "Audio streaming service exists", "weight": 0.3, "disconfirmingEvidence": ["HTTP probing returns 404 or generic HTML pages, indicating no active audio service."]}], "selected": 0}
```

Archive.org operates a large petabox CDN infrastructure

INFRASTRUCTURE **CONFIRMED** **PRIORITY: HIGH** - Investigation: archive.org

Subdomain enumeration returned dozens of hostnames containing the pattern "petabox" (e.g., ia-petabox-review-webdev-178-r40bhn.archive.org). These are known internal CDN nodes used by the Internet Archive to serve large static assets and Wayback Machine content.

```
{"technique": "QoIC", "inputs": [{"sourceId": "subfinder_domain:archive.org", "summary": "200 subdomains discovered, many with 'petabox' prefix indicating CDN nodes.", "credibility": 4, "recencyDays": 0}], "reasoning": "The presence of a consistent naming scheme across many subdomains strongly indicates a managed CDN infrastructure owned by the target. The pattern aligns with known Internet Archive petabox deployments.", "outputCandidates": [{"label": "Petabox CDN confirmed", "weight": 0.1, "disconfirmingEvidence": ["If the petabox subdomains resolve to unrelated third-party services or are not under Archive.org control."]}], "selected": 0}
```

Key Personnel Email Pattern

IDENTITY **PROBABLE** **PRIORITY: LOW** - Investigation: archive.org

Public staff emails at Internet Archive follow the pattern firstname.lastname@archive.org, e.g., brewster.kahle@archive.org.

```
{"technique": "Indicators", "inputs": [{"sourceId": "holehe:brewster.kahle@archive.org", "summary": "Holehe confirms email exists and is associated with Internet Archive staff.", "credibility": 3, "recencyDays": 180}], "reasoning": "Email pattern confirms organizational affiliation of founder.", "outputCandidates": [{"label": "Founder uses archive.org email", "weight": 0.3, "disconfirmingEvidence": ["No valid email found for founder at archive.org"]}], "selected": 0}
```

Domain Ownership

IDENTITY **PROBABLE** **PRIORITY: MEDIUM** - Investigation: archive.org

The domain archive.org is registered to the non-profit organization Internet Archive, Inc.

```
{"technique": "Indicators", "inputs": [{"sourceId": "whois:archive.org", "summary": "WHOIS record shows registrant organization Internet Archive, Inc.", "credibility": 4, "recencyDays": 30}], "reasoning": "Ownership aligns with the known organization behind the site.", "outputCandidates": [{"label": "Internet Archive, Inc. owns archive.org", "weight": 0.2, "disconfirmingEvidence": ["WHOIS showing a different registrant"]}], "selected": 0}
```

Internet Archive Founder

The Internet Archive (archive.org) was founded by Brewster Kahle in 1996.

```
{"technique": "ACH", "inputs": [{"sourceId": "https://en.wikipedia.org/wiki/Brewster_Kahle", "summary": "Wikipedia biography of Brewster Kahle stating he is the founder of the Internet Archive.", "credibility": 5, "recencyDays": 3650}], "reasoning": "Candidate identity: Brewster Kahle as founder. Disconfirming evidence would be any source naming a different founder. No such evidence found.", "outputCandidates": [{"label": "Brewster Kahle is founder", "weight": 0.1, "disconfirmingEvidence": ["Sources naming a different individual as founder"]}], "selected": 0}
```

Audit log (last 6 entries)

Forward-linked SHA-256 chain. Verifiable offline using the recipe in docs/07-SECURITY.md.

TIMESTAMP	ACTION	ENTITY	ACTOR	HASH
2026-05-20 20:24:48 UTC	bridge_investigation_to_case	Case cmpeigafg0000abuvfy8y320p	system	279cbd8d7924...
2026-05-20 20:24:48 UTC	findings_promoted_to_evidence	Case cmpeigafg0000abuvfy8y320p	system	3a0c23e585b1...
2026-05-20 20:24:48 UTC	generate_case_report	Case cmpeigafg0000abuvfy8y320p	Admin	96e520737088...
2026-05-20 20:25:55 UTC	generate_case_report	Case cmpeigafg0000abuvfy8y320p	Admin	2ffd716b1e86...
2026-05-20 20:27:53 UTC	generate_case_report	Case cmpeigafg0000abuvfy8y320p	Admin	2bce5d0f8d3a...
2026-05-20 20:32:49 UTC	generate_case_report	Case cmpeigafg0000abuvfy8y320p	Admin	d410ef2a1f4c...